This assignment will not be graded and is only for practice.

Level 1

1) Choose the correct options. **1 point**

- [ ] Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
- [ ] Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
- [ ] Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $< a, b >= 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv\rangle = \langle u, v\rangle$.
- [ ] Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $< a, b >= 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv\rangle = \langle u, v\rangle$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $< a, b >= 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv\rangle = \langle u, v\rangle$.

2) If A is the matrix representation of an orthogonal transformation $T : \mathbb{R}^n \to \mathbb{R}^n$, then choose the set of correct statements. **1 point**

- [ ] $AA^T = I$
- [ ] $A = A^T$
- [ ] AA^T is an upper triangular matrix
- [ ] AA^T is a lower triangular matrix
- [ ] $A^T = A^{-1}$
- [ ] $A^T = -A^{-1}$

- [ ] A^{-1} does not exist
- [ ] A^{-1} is also an orthogonal matrix

No, the answer is incorrect.

Score: 0

Accepted Answers:

$AA^T = I$
AA^T is an upper triangular matrix
AA^T is a lower triangular matrix
$A^T = A^{-1}$
A^{-1} is also an orthogonal matrix

3) If A is the matrix representation of an orthogonal transformation $T : \mathrm{R}^n \to \mathrm{R}^n$, then choose the set of correct statements. ***1 point***

- [ ] The column vectors of A are orthogonal but not orthonormal.
- [ ] The column vectors of A are orthonormal.
- [ ] The row vectors of A are orthogonal but not orthonormal.
- [ ] The row vectors of A are orthonormal.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The column vectors of A are orthonormal.
The row vectors of A are orthonormal.

Level 2

4) Consider the orthogonal set $B = \{(1,1,1), (-1,1,0)\}$ in $\mathbb{R}^3$. Find a third vector in $\mathbb{R}^3$ which is orthogonal to both the vectors in B. ***1 point***

- $(1, 1, -2)$
- $(1, -1, 2)$
- $(-1, 1, -2)$
- $(-1, -1, -2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 1, -2)$

5) Consider the orthogonal set $B = \{(1,1,1), (-1,1,0)\}$ and vector obtained in question 4 which is orthogonal to the set B. Form the columns of an orthogonal matrix U by scaling these vectors appropriately. Choose the correct option. ***1 point***

- $U = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$
- $U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$
- $U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$
- $U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$$

6) Consider a vector $v = (3, 0, 4)$ and U from Question 5. Choose the correct option using the dot product as the standard inner product. ***1 point***

- ○ $\|v\| = \|Uv\|$
- ○ $\|v\| = \|Uv\|$
- ○ $2\,\|v\| = \|Uv\|$
- ○ $\|v\| = 2\,\|Uv\|$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\|v\| = \|Uv\|$

7) Let the matrix representation of a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ be $U = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$. ***1 point***

Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$. Choose the set of correct statements.

- ☐ T is an orthogonal transformation.
- ☐ T is not an orthogonal transformation.
- ☐ B is a real symmetric matrix.
- ☐ $U^{-1}BU$ is a diagonal matrix.
- ☐ $U^{-1}BU$ is a scalar matrix.

- [ ] $U^{-1}BU$ is an upper triangular matrix.
- [ ] $U^{-1}BU$ is a lower triangular matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is an orthogonal transformation.
B is a real symmetric matrix.
$U^{-1}BU$ is a diagonal matrix.
$U^{-1}BU$ is an upper triangular matrix.
$U^{-1}BU$ is a lower triangular matrix.

Your score is: 0/7